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Original exam-style questions · 60 questions

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Part 1

0610_T16_P1_20261018

Part 1 · Question 1

core · 1 mark · 0610_T16_P1_20261018_Q1

One parent produces offspring without gametes. Apart from mutation, which feature is expected of the offspring?

A Each offspring has half the parent's genes.

B The offspring have the same genetic information as the parent.

C Every offspring has genes from two parents.

D All offspring must have different genes.

Part 1 · Question 2

core · 1 mark · 0610_T16_P1_20261018_Q2

A single-celled organism divides into two cells without fusion with another cell. Which process has occurred?

A pollination

B fertilisation

C sexual reproduction

D asexual reproduction

Part 1 · Question 3

supplement · 1 mark · 0610_T16_P1_20261018_Q3

A grower wants to multiply one plant with a particularly desirable fruit flavour. Which advantage most directly supports using asexual reproduction?

A The desirable genetic combination can be retained in the offspring.

B The offspring must have increased genetic variation.

C The offspring can only be formed after cross-pollination.

D Every offspring becomes resistant to all diseases.

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Part 1 · Question 4

core · 1 mark · 0610_T16_P1_20261018_Q4

Which event distinguishes sexual reproduction from asexual reproduction?

A increase in cell number

B development of offspring

C fusion of two gamete nuclei

D growth of a new organism

Part 1 · Question 5

core · 1 mark · 0610_T16_P1_20261018_Q5

A pollen grain lands on a stigma, a tube grows, and a pollen nucleus joins a nucleus in an ovule. Which event is fertilisation?

A the arrival of pollen on the stigma

B the opening of the flower

C the growth of the pollen tube alone

D the joining of the two nuclei

Part 1 · Question 6

supplement · 1 mark · 0610_T16_P1_20261018_Q6

In a species, each gamete nucleus has 12 chromosomes. How many chromosomes does the zygote nucleus have immediately after normal fertilisation?

A 6

B 24

C 12

D 48

Part 1 · Question 7

core · 1 mark · 0610_T16_P1_20261018_Q7

Which flower structure produces pollen grains?

A stigma

B ovary

C anther

D petal

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Part 1 · Question 8

core · 1 mark · 0610_T16_P1_20261018_Q8

Which pollen feature is most likely to help attachment to an insect's body?

A a spiky outer surface

B a very smooth surface that prevents attachment

C a low mass that allows prolonged wind transport

D a shape that prevents contact with insects

Part 1 · Question 9

core · 1 mark · 0610_T16_P1_20261018_Q9

During pollination, pollen grains move between which structures?

A stigma and ovule

B anther and stigma

C ovule and ovary

D petal and sepal

Part 1 · Question 10

core · 1 mark · 0610_T16_P1_20261018_Q10

Which combination is most typical of a wind-pollinated flower?

A nectar and strongly scented petals

B sticky pollen and brightly coloured petals

C exposed anthers and a large feathery stigma

D an enclosed stigma and very little pollen

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Part 1 · Question 11

core · 1 mark · 0610_T16_P1_20261018_Q11

Viable seeds are kept at a suitable temperature with oxygen available. Which treatment prevents germination because a required condition is absent?

A placing the seeds on moist cotton wool

B keeping the seeds moist in darkness

C supplying moist seeds with air

D keeping the seeds completely dry

Part 1 · Question 12

supplement · 1 mark · 0610_T16_P1_20261018_Q12

Pollen moves from one flower to another flower on the same plant. How is this classified?

A self-pollination

B cross-pollination

C asexual reproduction

D seed dispersal

Part 1 · Question 13

supplement · 1 mark · 0610_T16_P1_20261018_Q13

Which transfer is cross-pollination?

A pollen moving between two flowers on the same plant

B pollen moving within one flower

C pollen moving between flowers on two different plants of the same species

D seeds moving between two habitats

Part 1 · Question 14

supplement · 1 mark · 0610_T16_P1_20261018_Q14

A flowering plant species becomes more dependent on cross-pollination. Which potential disadvantage follows?

A Successful reproduction may depend more on pollen transfer between plants.

B Every offspring becomes genetically identical.

C Fertilisation no longer requires a pollen nucleus.

D Seeds no longer contain an embryo.

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Part 1 · Question 15

supplement · 1 mark · 0610_T16_P1_20261018_Q15

After a compatible pollen grain germinates, which route does the pollen tube take?

A through the sepal towards an anther

B through the style towards an ovule

C from the ovule to the petal

D from the filament into a leaf

Part 1 · Question 16

supplement · 4 marks · 0610_T16_P1_20261018_Q16

Two fields contain plants of one crop species. Field X contains clones of one parent, while field Y contains sexually produced plants.

(a) Explain one advantage of the plants in field X for harvesting a uniform crop. [2]

(b) Explain one risk to field X if a new infectious disease arrives. [2]

Part 1 · Question 17

supplement · 4 marks · 0610_T16_P1_20261018_Q17

A rare plant grows alone in a newly formed habitat where there are no other plants of its species.

(a) Explain why asexual reproduction can allow its population to increase quickly. [2]

(b) Explain why a population produced this way may respond poorly to a major environmental change. [2]

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Part 1 · Question 18

core · 4 marks · 0610_T16_P1_20261018_Q18

Two gametes fuse and the resulting cell develops into an offspring.

(a) Describe fertilisation and identify the resulting cell. [2]

(b) Describe two genetic features expected of offspring produced sexually. [2]

Part 1 · Question 19

supplement · 4 marks · 0610_T16_P1_20261018_Q19

A species has 18 chromosomes in a normal body-cell nucleus.

(a) State the chromosome number in a gamete and explain your answer. [2]

(b) State the chromosome number in the zygote and explain how it is restored. [2]

Part 1 · Question 20

supplement · 4 marks · 0610_T16_P1_20261018_Q20

A crop breeder uses sexual reproduction to produce plants for a region with a changing climate.

(a) Explain one potential advantage of this method. [2]

(b) Explain two disadvantages for producing a uniform crop rapidly. [2]

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Part 1 · Question 21

supplement · 4 marks · 0610_T16_P1_20261018_Q21

A wild animal population produces genetically different offspring sexually.

(a) Explain how variation may help when a new predator arrives. [2]

(b) Explain why sexual reproduction can fail when suitable individuals become widely separated. [2]

Part 1 · Question 22

core · 4 marks · 0610_T16_P1_20261018_Q22

A flower has a damaged stigma but intact anthers and ovary.

(a) State the functions of the stigma and the ovary. [2]

(b) Suggest why this damage could reduce seed production without reducing pollen production. [2]

Part 1 · Question 23

core · 4 marks · 0610_T16_P1_20261018_Q23

Flower A is brightly coloured and produces nectar. Flower B has small dull petals and exposed hanging anthers.

(a) Identify the likely pollinating agent of each flower and give one reason for each identification. [2]

(b) Describe two differences expected between their pollen grains. [2]

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Part 1 · Question 24

core · 4 marks · 0610_T16_P1_20261018_Q24

Pollen reaches a flower but no seed subsequently develops.

(a) Distinguish pollination from fertilisation. [2]

(b) Suggest two reasons why pollen arrival might not lead to fertilisation. [2]

Part 1 · Question 25

core · 4 marks · 0610_T16_P1_20261018_Q25

A student compares equal numbers of viable seeds from the same species. Set A is moist, aerated and at 22 °C. Set B is dry, aerated and at 22 °C.

(a) Identify the condition being tested and state the expected difference in germination. [2]

(b) State two conditions that must be kept the same to make this comparison valid. [2]

Part 1 · Question 26

core · 4 marks · 0610_T16_P1_20261018_Q26

A student places moist seeds in two containers at the same suitable temperature. One contains air; the other contains an oxygen-free gas.

(a) State the expected result and the conclusion if only seeds supplied with air germinate. [2]

(b) Suggest two further controls needed for a fair comparison. [2]

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Part 1 · Question 27

supplement · 4 marks · 0610_T16_P1_20261018_Q27

Pollen is transferred from flower X to flower Y on the same plant, and from flower X to flower Z on another plant of the same species.

(a) Classify each transfer and justify the classification. [2]

(b) Describe the starting and receiving structures in both transfers. [2]

Part 1 · Question 28

supplement · 4 marks · 0610_T16_P1_20261018_Q28

Two populations of the same plant species differ in their usual pollination pattern. Population P mostly self-pollinates; population Q mostly cross-pollinates.

(a) Explain which population is more likely to produce a wider range of inherited combinations. [2]

(b) Explain how the populations might differ if pollinating insects become scarce. [2]

Part 1 · Question 29

supplement · 4 marks · 0610_T16_P1_20261018_Q29

A compatible pollen grain is present on a receptive stigma.

(a) Describe the route by which its male nucleus reaches a nucleus in an ovule. [2]

(b) Describe the nuclear event that follows and state its product. [2]

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Part 1 · Question 30

core · 4 marks · 0610_T16_P1_20261018_Q30

A wind-pollinated grass produces many pollen grains. Its mature seeds are later stored dry before sowing.

(a) Explain how two flower features can increase the chance of wind pollination. [2]

(b) State two conditions, other than adding water, needed for the stored viable seeds to germinate. [2]

Part 2

0610_T16_P2_20261018

Part 2 · Question 1

core · 1 mark · 0610_T16_P2_20261018_Q1

Which sperm structure releases substances that help it pass through the coverings around an egg?

A flagellum

B acrosome

C nucleus alone

D cell membrane alone

Part 2 · Question 2

core · 1 mark · 0610_T16_P2_20261018_Q2

Immediately after fertilisation, which change helps prevent further sperm entering an egg?

A loss of all stored nutrients

B formation of a second egg nucleus

C growth of a flagellum

D a change in its jelly coat

Part 2 · Question 3

core · 1 mark · 0610_T16_P2_20261018_Q3

Which comparison between human gametes is correct?

A Egg cells are larger than sperm cells.

B Egg cells normally swim using a flagellum.

C Sperm cells contain larger nutrient stores than egg cells.

D Egg cells are produced in greater numbers than sperm cells.

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Part 2 · Question 4

core · 1 mark · 0610_T16_P2_20261018_Q4

After fertilisation, which sequence describes early development correctly?

A embryo, zygote, implantation in the uterus lining

B zygote, implantation in an anther, embryo

C zygote, embryo, implantation in the uterus lining

D embryo, implantation in the ovary, zygote

Part 2 · Question 5

supplement · 1 mark · 0610_T16_P2_20261018_Q5

Which pair normally passes from fetal blood towards maternal blood at the placenta?

A oxygen and glucose

B glucose and amino acids

C oxygen and amino acids

D carbon dioxide and urea

Part 2 · Question 6

supplement · 1 mark · 0610_T16_P2_20261018_Q6

Which statement about the placenta is correct?

A It blocks every harmful substance.

B Some toxins and pathogens can cross it.

C It prevents all exchange of dissolved substances.

D It converts fetal blood into maternal blood.

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Part 2 · Question 7

core · 1 mark · 0610_T16_P2_20261018_Q7

Which hormone is mainly responsible for development of male secondary sexual characteristics at puberty?

A oestrogen

B FSH alone

C testosterone

D progesterone

Part 2 · Question 8

core · 1 mark · 0610_T16_P2_20261018_Q8

What happens to the uterus lining during menstruation?

A It breaks down and is shed.

B It becomes a pollen tube.

C It releases the egg from an ovary.

D It is converted into sperm cells.

Part 2 · Question 9

supplement · 1 mark · 0610_T16_P2_20261018_Q9

Which hormone directly triggers ovulation in the menstrual cycle?

A testosterone

B LH

C insulin

D adrenaline

Part 2 · Question 10

supplement · 1 mark · 0610_T16_P2_20261018_Q10

Which structure normally produces progesterone after ovulation and before pregnancy is established?

A pituitary gland

B oviduct lining

C corpus luteum

D urinary bladder

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Part 2 · Question 11

core · 1 mark · 0610_T16_P2_20261018_Q11

Which description defines a sexually transmitted infection?

A any inherited disorder

B any illness caused by poor nutrition

C an infection transmitted only by coughing

D an infection transmitted through sexual contact

Part 2 · Question 12

core · 1 mark · 0610_T16_P2_20261018_Q12

Which statement correctly links HIV and AIDS?

A HIV infection may lead to AIDS.

B AIDS is the organism that produces HIV.

C HIV is a bacterium that always causes immediate AIDS.

D Every STI is caused by HIV.

Part 2 · Question 13

core · 1 mark · 0610_T16_P2_20261018_Q13

Which situation can transmit HIV?

A sharing a clean classroom desk

B shaking hands with an infected person

C sharing a needle contaminated with infected blood

D standing beside an infected person

Part 2 · Question 14

core · 1 mark · 0610_T16_P2_20261018_Q14

Which action directly reduces the risk of sexual transmission of HIV?

A correct use of a condom

B taking antibiotics before every sexual contact

C sharing injection equipment

D washing hands instead of using protection

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Part 2 · Question 15

core · 1 mark · 0610_T16_P2_20261018_Q15

Which transmission route can infect an infant with HIV?

A inheritance of an HIV allele from an uninfected parent

B transfer from an infected mother during breastfeeding

C breathing the same air as an infected adult

D receiving a clean blanket

Part 2 · Question 16

core · 4 marks · 0610_T16_P2_20261018_Q16

A sperm cell reaches an egg cell.

(a) Describe fertilisation and state the cell formed. [2]

(b) Explain how a flagellum and mitochondria help a sperm reach an egg. [2]

Part 2 · Question 17

core · 4 marks · 0610_T16_P2_20261018_Q17

Human eggs and sperm are specialised for different roles.

(a) Explain why an egg has a relatively large energy store. [2]

(b) Compare the motility and typical numbers of the two gametes. [2]

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Part 2 · Question 18

core · 4 marks · 0610_T16_P2_20261018_Q18

A fertilised egg develops before implantation.

(a) Describe the change from zygote to early embryo. [2]

(b) State the site of implantation and describe what happens there. [2]

Part 2 · Question 19

supplement · 4 marks · 0610_T16_P2_20261018_Q19

Placental exchange supplies a growing fetus.

(a) State two dissolved nutrients transferred from maternal blood to fetal blood. [2]

(b) State two excretory products transferred in the opposite direction. [2]

Part 2 · Question 20

supplement · 4 marks · 0610_T16_P2_20261018_Q20

A pregnant woman is exposed to a toxin that can pass through the placenta.

(a) Describe the role of the placenta and umbilical cord in supplying fetal tissues. [2]

(b) Explain why the placenta does not guarantee protection from this exposure. [2]

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Part 2 · Question 21

core · 4 marks · 0610_T16_P2_20261018_Q21

Some sperm have a damaged acrosome, while an egg has normal stored nutrients.

(a) Explain why the damaged sperm may be less successful at fertilisation. [2]

(b) Explain two functions of adaptations of the egg. [2]

Part 2 · Question 22

core · 4 marks · 0610_T16_P2_20261018_Q22

A researcher compares sperm and egg cells without using pictures.

(a) Describe two structural differences. [2]

(b) Explain how their numbers and motility differ. [2]

Part 2 · Question 23

core · 4 marks · 0610_T16_P2_20261018_Q23

Two adolescents develop secondary sexual characteristics during puberty.

(a) State the hormone mainly responsible for male changes and give one example of such a change. [2]

(b) State the hormone mainly responsible for female changes and give one example. [2]

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Part 2 · Question 24

core · 4 marks · 0610_T16_P2_20261018_Q24

A woman has a regular 28-day menstrual cycle and does not become pregnant.

(a) Describe the events in the ovary around the middle of the cycle. [2]

(b) Describe changes in the uterine lining before and at the next menstruation. [2]

Part 2 · Question 25

supplement · 4 marks · 0610_T16_P2_20261018_Q25

After ovulation, progesterone concentration rises.

(a) State the ovarian structure producing progesterone and describe its effect on the uterine lining. [2]

(b) Explain how high progesterone helps prevent another ovulation in that cycle. [2]

Part 2 · Question 26

supplement · 4 marks · 0610_T16_P2_20261018_Q26

A developing ovarian follicle releases increasing amounts of oestrogen.

(a) Describe two roles of FSH early in the menstrual cycle. [2]

(b) Describe two effects associated with increasing oestrogen before ovulation. [2]

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Part 2 · Question 27

core · 4 marks · 0610_T16_P2_20261018_Q27

A school provides education about sexually transmitted infections.

(a) Describe what is meant by an STI and give one example of an STI pathogen. [2]

(b) Explain how education and condom use can reduce transmission. [2]

Part 2 · Question 28

core · 4 marks · 0610_T16_P2_20261018_Q28

A clinic introduces screened blood donations and single-use injection needles.

(a) Explain how each measure reduces HIV transmission. [2]

(b) State two ways HIV may pass from an infected mother to her infant. [2]

Part 2 · Question 29

core · 4 marks · 0610_T16_P2_20261018_Q29

A public information leaflet incorrectly calls AIDS a bacterium.

(a) State what HIV is and describe its relationship with AIDS. [2]

(b) Explain why contact with an infected person does not always mean HIV transmission. [2]

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Part 2 · Question 30

core · 4 marks · 0610_T16_P2_20261018_Q30

A person uses shared injection equipment and has unprotected sexual contact.

(a) Identify the HIV transmission risk associated with each behaviour. [2]

(b) State one specific action to reduce each risk. [2]